Natural Language Processing Journal 9 (2024) 100104

Contents lists available at ScienceDirect

Natural Language Processing Journal

journal homepage: www.elsevier.com/locate/nlp

Cyberbullying detection of resource constrained language from social media using transformer-based approach

Syed Sihab-Us-Sakib [a], Md. Rashadur Rahman [a,*], Md. Shafiul Alam Forhad [a], Md. Atiq Aziz [b]

[a] Department of Computer Science and Engineering, Chittagong University of Engineering and Technology, Chattogram, 4349, Bangladesh
[b] Department of Computer Science and Engineering, Bangladesh Army University of Science and Technology, Saidpur, 5311, Bangladesh

ARTICLE INFO

Keywords:
Cyberbullying classification
Social media
Transformers
Machine Learning
Low resource language
Natural Language Processing

ABSTRACT

The rise of the internet and social media has facilitated diverse interactions among individuals, but it has also led to an increase in cyberbullying—a phenomenon with detrimental effects on mental health, including the potential to induce suicidal thoughts. To combat this issue, we have developed the Cyberbullying Bengali Dataset (CBD), a novel resource containing 2751 manually labeled texts categorized into five classes, including various forms of cyberbullying and non-bullying instances. In our study on cyberbullying detection, we conducted an extensive evaluation of various machine learning and deep learning models. Specifically, we examined Support Vector Machine (SVM), Multinomial Naive Bayes (MNB), and Random Forest (RF) among the traditional machine learning models. For deep learning models, we explored Gated Recurrent Unit (GRU), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM). We have also experimented with state-of-the-art transformer architectures, including m-BERT, BanglaBERT, and XLM-RoBERTa. After rigorous experimentation, XLM-RoBERTa emerged as the most effective model, achieving a significant F1-score of 0.83 and an accuracy of 82.61%, outperforming all other models. Our work provides insights into effective cyberbullying detection on platforms like Facebook, YouTube, and Instagram.

1. Introduction

In recent years, the rapid advancement of electronic technologies and the widespread use of the internet have brought about a pressing societal concern: cyberbullying. The term "cyberbullying" was first coined around 2000 in Canada by a website owner focused on traditional bullying prevention (Aboujaoude et al., 2015). Tokunaga provided an early definition, describing cyberbullying as "any behavior conducted through electronic or digital means by individuals or groups that repeatedly transmit hostile or aggressive messages with the intent to harm or discomfort others" (Tokunaga, 2010). This definition elucidates the technological aspect, the presence of hostility, the crucial intent to inflict harm—a consensus among scholars and the repetitive nature of cyberbullying (Aboujaoude et al., 2015). Recognizing these fundamental features is imperative for formulating effective strategies to mitigate the detrimental effects of cyberbullying on individuals and communities.

Research indicates that a staggering 25% of internet users have encountered some form of cyberbullying (Hemmatian and Sohrabi, 2019). This pervasive issue transcends age barriers, affecting individuals across all demographics, with teenagers being particularly susceptible in the realm of online social networks. The primary objective of cyberbullying is to undermine the victim's self-esteem and social status, often resulting in feelings of despair and isolation. Tragically, in severe cases, cyberbullying has led to tragic outcomes, including loss of life for the victim (Hellfeldt et al., 2020). Ideological disparities frequently serve as catalysts for hatred and the use of abusive language, exacerbating instances of cyberbullying. The increasing occurrence of offensive language across various text-based online platforms like Facebook, Instagram, and YouTube highlights the urgent necessity for thorough analysis and intervention to address this pervasive problem. It is important to note that cyberbullying predominantly manifests through text-based communication (Dewani et al., 2021), taking various forms such as racism, flaming, threats, or body shaming. Such malicious behavior can compel individuals to withdraw and isolate themselves from online interactions.

Social media platforms facilitate dynamic exchanges of information and ideas but are plagued by cyberbullying, a pervasive issue (Chakraborty and Seddiqui, 2019). Facebook, with billions of users, serves as a prominent hub for communication, yet discussions often turn hostile, targeting individuals or groups (Ishmam and Sharmin, 2019). The incidence of cyberbullying is witnessing an upward trend

* Corresponding author.
E-mail addresses: sakibshihab12@gmail.com (S. Sihab-Us-Sakib), rashadur@cuet.ac.bd (M.R. Rahman), forhad@cuet.ac.bd (M.S.A. Forhad), atiqaziz.bd@gmail.com (M.A. Aziz).

https://doi.org/10.1016/j.nlp.2024.100104
Received 23 May 2024; Received in revised form 26 August 2024; Accepted 8 September 2024

2949-7191/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

alongside the continuous expansion of social media platforms (Kee et al., 2022). Within these networks, individuals often experience harassment from strangers and unauthorized users (Das et al., 2019).

We examine cyberbullying detection methods to understand existing approaches and suggest future research directions. Through a comprehensive literature review, we address the challenges and opportunities in combating this societal issue. In the field of Natural Language Processing (NLP), detecting cyberbullying is particularly challenging due to the absence of clear online etiquette.

Celebrities, women, and leaders often experience threatening language, contributing to high instances of bullying, notably in regions like Pakistan (Khan et al., 2022). Talpur et al. (2020) recently provided an overview of models excelling in identifying cyberbullying across diverse social media platforms. However, research in this area has primarily focused on English and Twitter datasets, leaving a critical gap in Bengali cyberbullying detection. The complexity of Bengali, with its low-resource status and intricate grammar, poses unique challenges.

Efforts to address this gap encompass the creation of specialized Bengali cyberbullying datasets and the utilization of machine learning, deep learning, and transformer-based techniques. Notably, these methods have primarily been employed for multiclass classification tasks, as demonstrated in the study by Das et al. (2021). Such endeavors highlight the necessity of formulating efficient strategies to detect and mitigate cyberbullying, especially in languages other than English, to foster safer online environments.

Our research addresses these challenges through the following key contributions:

- **Dataset:** Development of a new comprehensive Bengali cyberbullying corpus consisting of 2751 texts among which 2164 cyberbullying and 587 are non-bullying texts. Furthermore, by employing our annotation schema, cyberbullying texts are annotated into Cy-Flaming, Cy-Threat, Cy-Pull-a-Pig, Cy-Racism which will serve as a foundational resource for future investigations in this area
- **Comprehension:** Give insightful analysis and thorough data statistics to guarantee the dataset's quality
- **Model:** Tuned XLM-R identifies and categorizes cyberbullying Bengali texts highlighting the softmax probability of the participating classifiers according to their past performance on the dataset. On the developed dataset, our proposed model performs better than other models
- **Standardization:** Establish a benchmark work for future comparisons by examining and contrasting the suggested model's performance with other ML, DL baselines, and current methodologies
- **Error analysis:** Deeply examine the suggested model's outcomes and inaccuracies. provides both qualitative and quantitative analysis that clarifies the causes of some of the errors and offers some recommendations for how to improve the system's shortcomings

The paper's structure is outlined as follows: Section 2 presents related works on cyberbullying detection from social media. Section 3 describes the steps involved in preparing the dataset, including data collection and the labeling process and the overall methodology of our system is detailed, encompassing data preprocessing, feature extraction, and more. Section 4 outlines the applied machine learning, deep learning, and transformer model algorithms, along with their corresponding results. Subsequently, Section 5 delves into the conclusions drawn from the study's findings and delineates avenues for future research.

2. Related work

Numerous studies have delved into cyberbullying detection underscoring the imperative for effective strategies in combating cyberbullying across social media platforms. Cyberbullying, prevalent on platforms like Twitter and Facebook, significantly impacts the well-being of individuals, particularly the younger demographic (Van Hee et al., 2018). Such behavior, encompassing harassment, threats, and humiliation, exacts a toll on emotional and psychological health. The task of identifying cyberbullying is intricate, demanding consideration of factors such as language usage and message dynamics (Kee et al., 2022; Dong et al., 2020).

For detecting cyberbullying, the majority of work has been conducted in English due to the availability of datasets. In Khan et al. (2022) detected cyber aggression using a cyber-troll dataset for binary classification. They employed various ML models such as SVM, NB, and deep learning models like KNN and CNN. The most accurate result was obtained from their proposed DNN model with three layers, achieving 88.34% accuracy. A. Dewani et al. (2021) performed binary classification to detect cyberbullying in the resource-constrained language Urdu from Twitter data, achieving the highest accuracy of 85% using RNN-LSTM. Chelmis and Yao (2019) assessed cyberbullying severity from Instagram data, resulting in a low resource effect with the highest accuracy of 81.57% for the VLFH model. Bozyiğit et al. (2021) conducted binary classification to detect cyberbullying using SVM, LR, KNN, RF, and AdaBoost, achieving the highest accuracy of 90.10%. They used a corpus collected from tweets, where RF gave the highest accuracy of 91.153% (Talpur and O'Sullivan, 2020). Lu et al. (2020) proposed a new method called Char-CNNS to feed the neural network at the character level to identify cyberbullying. The study in Bello et al. (2023) combined six datasets from Kaggle and implemented CNN, RNN, BiLSTM and also implemented these models with the Word2Vec approach. But the transformer-based model BERT with CNN, BERT with RNN, and BERT with BiLSTM yielded the best results which gave an accuracy of 93% for each of the cases which indicates that BERT-based approaches are more effective in capturing the context of sentiment analysis tasks than traditional and Word2Vec approaches.

Numerous studies have explored text categorization in various languages, including English and Bengali, to detect offensive content (Mahmud et al., 2022; Ahmed et al., 2022; Emon et al., 2022; Shakambhari et al., 2022; Kumar and Sachdeva, 2022). Bengali, ranked seventh among the top 100 most spoken languages globally, serves as a primary language for over 210 million people, primarily residing in Indian states like West Bengal, Assam, and Tripura, as well as in immigrant communities in the UK, US, and the Middle East (Sen et al., 2022). Detecting cyberbullying in Bengali is crucial due to its widespread use. Despite the existence of various approaches in the literature for detecting Bengali cyberbullying (Ahmed et al., 2022; Mahmud et al., 2022; Emon et al., 2022; Aurpa et al., 2022), their performance in bullying detection remains unsatisfactory. Consequently, there is a significant need for an improved strategy capable of identifying different types of abusive Bengali content. Transformer-based approaches offer promise in effectively identifying and removing such content from social media platforms.

Although Bengali ranks as the seventh most spoken language globally, research on cyberbullying detection in this language is limited. Faisal Ahmed et al. (2021) proposed a cyberbullying detection framework involving two levels: first, determining if a text constitutes cyberbullying or not, and second, multiclass classification for cyberbullying in Bengali corpora. Ahmed et al. (2021) utilized three corpora, including one for Bengali, one for romanized Bengali, and a combination of both, achieving the highest accuracy of 84%.

In Akhter et al. (2023), they introduce a hybrid ML model tailored specifically for Bengali cyberbullying detection on social media platforms. Their approach involves preprocessing the text using the TfidfVectorizer (TFID) to transform Bengali text data into a useful format. The results presented in Akhter et al. (2023) demonstrate significant success, with the model achieving high accuracy rates of 98.57% and 98.82% in binary and multilabel classification, respectively. This performance surpasses previous benchmarks in Bengali cyberbullying detection. However, the study acknowledges limitations such as language specificity and dataset size constraints. In their future research direction, the authors aim to explore advanced techniques

such as deep learning and transformer-based approaches to further enhance cyberbullying detection. They suggest that models like BERT and RoBERTa could be particularly beneficial for improving the accuracy and adaptability of detection systems across different languages and cultural contexts. In Islam et al. (2022), they evaluate various machine learning and deep learning models for predicting suicide attempts using BanglaSPD. Notably, a deep learning model integrating CNN and BiLSTM achieves a noteworthy F1-Score of 0.61, indicating its effectiveness in identifying suicidal content in Bangla. Overall, their research provides significant resources and insights for suicide prevention efforts, especially in linguistically diverse contexts. In the study conducted by Abdullah et al. (2023), a Bengali dataset focusing on political aggression is introduced. The dataset categorizes texts into aggressive or non-aggressive, with further subcategories for aggressive texts such as inciting assassination, rebellion, or communal violence. Leveraging advanced BERT transformer models like Bangla-BERT and Multilingual-BERT, they achieve a notable 0.87 F1 score. This research significantly contributes to the comprehension of political aggression in online discourse and provides valuable insights for addressing its real-world impact.

In Aurpa et al. (2022), the authors delve into the challenges posed by the rise of user-generated content on platforms like Facebook, specifically in addressing abusive comments such as threats and harassment. Despite the increasing presence of Bangla content on Facebook, research on detecting abusive Bangla comments remains limited. Their study aims to swiftly identify and filter out such comments using transformer-based deep neural network models. Leveraging pre-trained language architectures like BERT and ELECTRA, they analyze a new dataset of 44,001 Facebook comments, achieving notable test accuracies of 85.00% and 84.92%, respectively.

The aforementioned studies collectively provide a compelling justification for our current research focus on cyberbullying detection in Bengali text. They underscore the efficacy of various ML, DL, and transformer-based models in identifying and categorizing abusive and bullying content. However, these studies exhibit certain limitations. To bridge these gaps, our study aims to develop a transformer-based approach tailored to Bengali languages. We will explore a range of ML, DL, and Transformer-based algorithms, providing detailed justifications for our classifications and contributing to the advancement of cyberbullying detection strategies. In this context, we use XLM-RoBERTa, a multilingual version of RoBERTa for cyberbullying detection from social media. We emphasize its superiority over traditional machine learning approaches and deep learning methods like BiLSTM. XLM-RoBERTa outperforms other models in Bangla text classification, particularly in multiclass cyberbullying detection, owing to its multilingual pretraining, robust representation learning, fine-tuning capability, and track record of achieving state-of-the-art results.

3. Methodology

In this section, we present our proposed methodology along with the different machine learning, deep learning, and transformer-based algorithms employed in the framework. Firstly, we will elucidate the functioning of the proposal. Subsequently, we will provide a succinct description of the machine learning models.

Fig. 1 illustrates the workflow of our proposal, which comprises six key components: Bengali cyberbullying text data collection, data annotation, text preprocessing for data preparation, feature extraction to convert text data into numerical information, application of ML, DL, and transformer algorithms for model building, and finally, performance evaluation using various metrics. We will now briefly outline the workflow of each component as follows:

3.1. Dataset construction

In this subsection, we elaborate on the process of data collection for our study on cyberbullying in Bengali which is depicted in Fig. 2. We had to construct a corpus comprising Bengali texts, following the guidelines provided by Vidgen and Derczynski (2020). The dataset construction involved the following steps:

- **Data collection:** We primarily sourced data from Facebook,[1,2] YouTube,[3] and Instagram.[4] Most of the bullying text was found in the comment sections of posts made by celebrities and viral news. We manually collected 2751 texts. The acquisition summary is illustrated in Fig. 3.
- **Data cleaning:** The following measures are taken to clean our collected text data for efficient training of our model:
  Removing unusual-size text: We filtered out sentences with fewer than three words, resulting in the exclusion of 118 texts.
  Unicode removal: We removed all characters not in the Bengali script Unicode range, including symbols, question marks, emojis, and characters from other languages. Specifically, we retained only characters within the Bengali Unicode range (0980-09FF) and removed Latin characters, numbers, and other scripts. To ensure consistency, we have standardized similar characters.
  Punctuation removal: We also removed irrelevant punctuation while keeping relevant Bengali punctuation. For numbers, we considered whether to keep, remove, or convert them to words. Furthermore, we stripped leading and trailing spaces and normalized multiple spaces to a single space. Special characters, such as emojis and symbols, were either removed or replaced as needed.
- **Annotation:** The collected texts were annotated for various types of cyberbullying behaviors. Numerous studies emphasized the importance of selecting annotators with a diverse range of opinions and experiences. Additionally, they recommend making the annotators' identities clear (Bender and Friedman, 2018). A summary of the annotators' demographic information is provided in Table 1.

Furthermore, we provide definitions of cyberbullying behaviors, outlined as follows:

Cyberbullying: The practice of humiliating or threatening others through digital and technological tools such as mobile phones, chat rooms, emails, and OSNs platforms like Facebook and Twitter Talpur et al. (2020).

Cy-flaming: Two or more internet users attacking one another is known as flaming Willard (2007).

Cy-threat: Involves sending text messages including threats to another online user or groups, similar to cyberstalking Talpur et al. (2020).

Cy-racism: Discrimination based on a person's race, nationality, or religion Pascucci et al. (2019).

Cy-pull-a-pig: Incidents where young people deemed obese and ugly are victims Pascucci et al. (2019).

Not bullying: The absence of any action or content that could be seen as cyberbullying Talpur et al. (2020).

The annotation procedure, as depicted in Fig. 4, adhered to standard guidelines (Ross et al., 2017; Fortuna et al., 2019). Texts were examined to determine if they contained instances of cyberbullying. Those identified as such were included in the dataset and subjected to further annotation based on the definitions which is elaborated above. In cases where annotators disagreed, an expert was available to arbitrate. The disputed text might either be excluded or incorporated into the dataset based on its validity, as determined by the expert.

[1] www.facebook.com/therealsalmanmuqtadir
[2] www.facebook.com/pori.monii
[3] www.youtube.com/@heroalomofficialchannel
[4] www.instagram.com/pori.moni.902

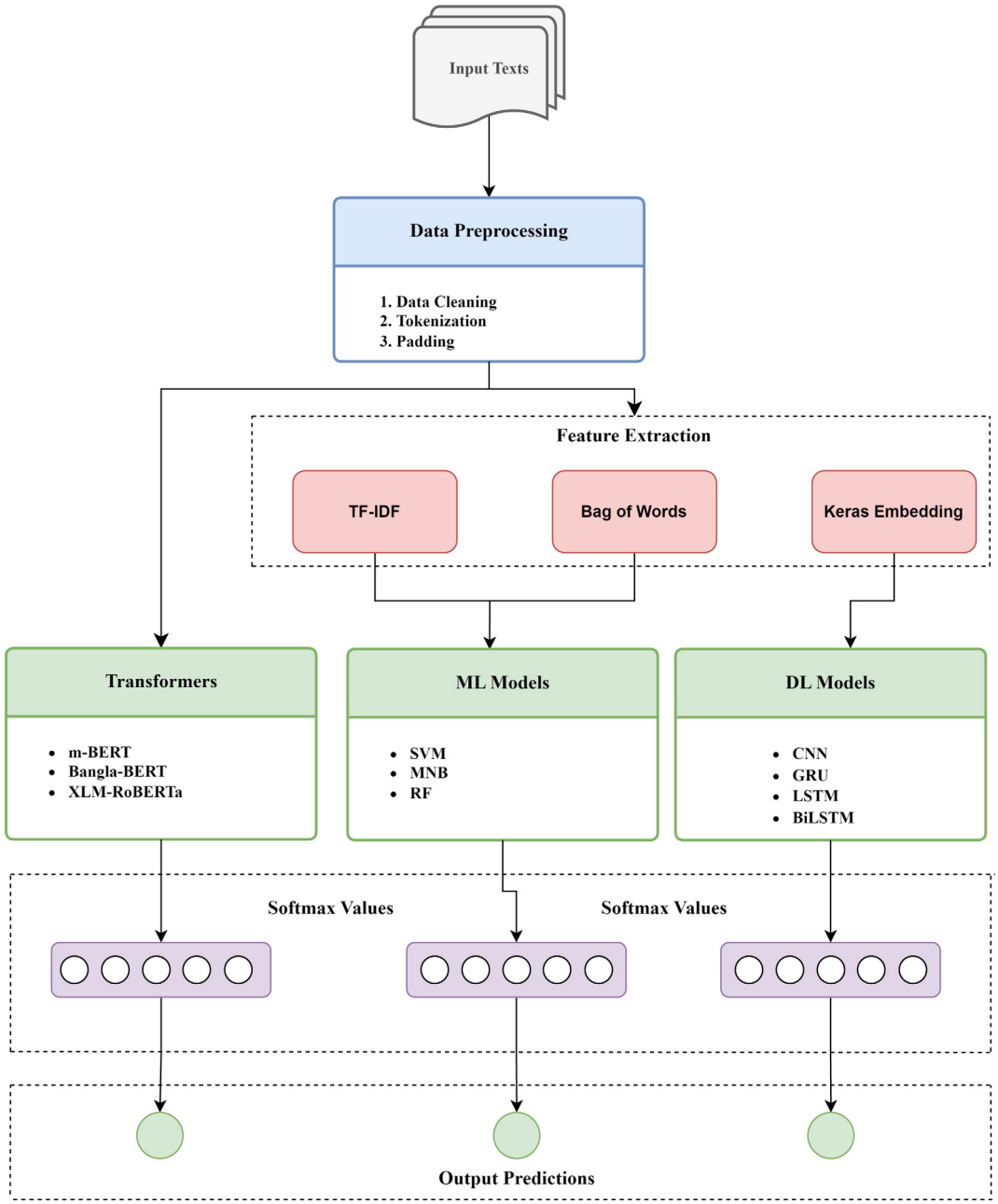

Fig. 1. Abstract overview of methodology of our proposed work.

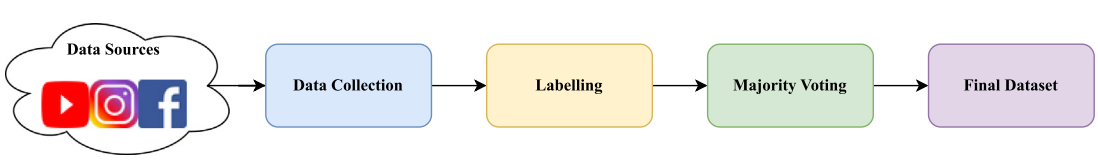

Fig. 2. Dataset development steps.

Two annotators have annotated the dataset and when there is any disagreement the expert has resolved it and the whole process is supervised by the expert. We calculated the degree of agreement using Cohens Kappa's reliability measure. The measurement of agreement throughout different classes is shown in Fig. 5 where the average measurement of agreement is 0.88 which indicates almost perfect agreement.

At last, the annotation and a sample of our dataset are provided in Table 2, which displays text samples for each of our classes. On the other hand, Fig. 8 illustrates the document summary of our dataset, providing information on the number of documents, words, and unique words available for each class. The length distribution is depicted in Fig. 9, indicating the frequency of different lengths in our dataset.

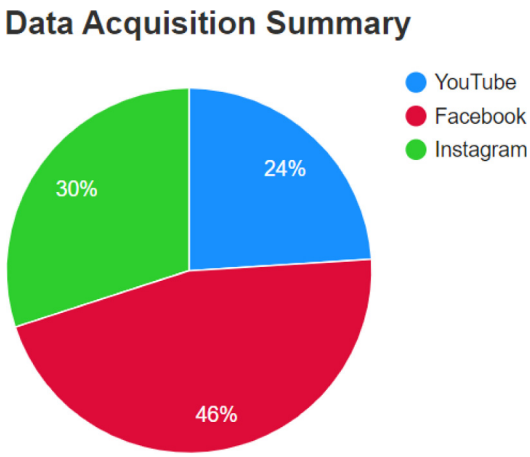

Fig. 3. Data acquisition summary.

Table 1
Demographic information of annotators.

Annotator features	AN1	AN2	Expert
Delegation	Undergrad Student	Fresh Graduate	Assistant Professor
Research Field	NLP	NLP	NLP
Age	22–25	22–25	25–40
Experience	1 Year	2 Year	4 Years

3.2. Dataset summary

There are a total of 2751 texts in our dataset which were classified according to our classification schema Fig. 6. Among these, 632 texts are classified as Cy-Flaming, 500 texts as Cy-Threat, 580 texts as Cy-Racism, 452 texts as Cy-Pull-a-Pig, and 587 texts as Not Bullying, as illustrated in Fig. 7.

3.3. Data preprocessing

Data must be preprocessed before being fed into the models. This subsection outlines the various steps involved in data preprocessing:

- **Data cleaning:** Removes unwanted symbols, stopwords, low-length data, and punctuation using either regular expressions.
- **Tokenization:** Converts preprocessed text into a vector of distinct numbers using the Keras tokenizer function.
- **Padding:** Creates equal-length vectors and a vectorized padded text sequence for model input.
- **Normalization:** Transforms data into a common scale to speed up training and reduce the chances of getting stuck in local optima.

3.4. Feature extraction

- **TF-IDF:** Each term in a document is weighted based on its frequency within that document and its inverse frequency across the entire corpus. In our experiment, we established a vector size of 1065 based on the unique words present in our corpus.
- **Bag-of-words:** Converts text documents into a matrix format, where each row represents a document and each column represents a phrase or word. The matrix contains frequency counts of terms in each document.
- **Keras embedding:** Represents words as dense vectors in a high-dimensional space to capture semantic links and meaning. Implemented with Keras, using an input dimension of 6308, an output dimension of 64, and a maximum input length of 70.

3.5. Classification models

All the classification models have been developed in Python where Pandas and NumPy are used to prepare the data and 'Scikit-learn' is used for implementing the models. In machine learning algorithms, the random state is set to 95 for randomness involved in splitting data, random sampling, and model initialization. 'Gini' is selected for the criterion of RF and the 'Linear' kernel is selected for matching the similarity with an original feature in SVM. In deep learning models, epoch sizes of 50, 15, 35, and 35 are used for GRU, CNN, LSTM, and BiLSTM to allow the models to learn iteratively. A batch size of 32 is used for GRU and 64 is used for CNN, LSTM and BiLSTM which allows for determining the number of training samples to be processed before the models' parameters are updated. ReLU activation is used for GRU which solves the vanishing gradient problem in the hidden layer and softmax activation is used for CNN, LSTM, and BiLSTM which enables probability distribution for multi class classification. In transformers,

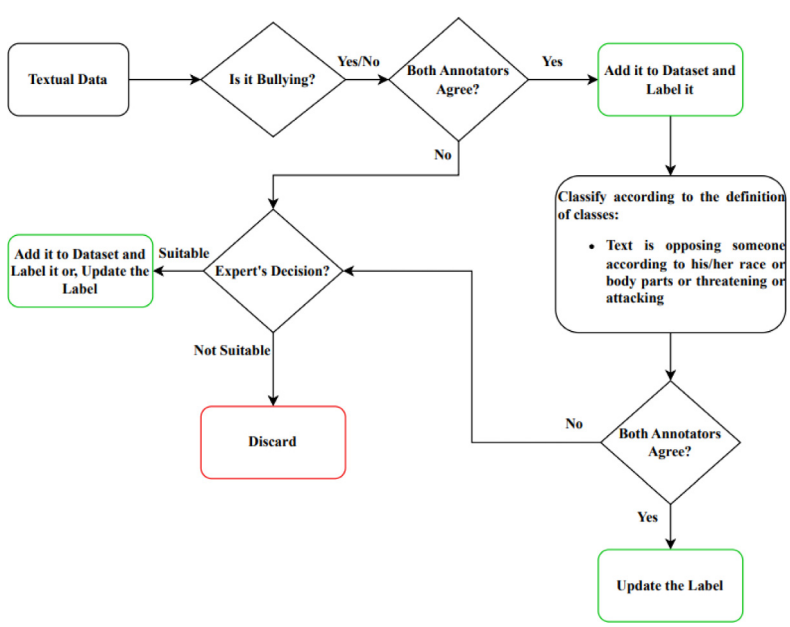

Fig. 4. Annotation procedure.

Table 2
Sample dataset.

Serial No.	Bangla Texts/Comments	Category
1	আপনি একটা অসুস্থ মানসিকতার লোক (You're a mentally ill person)	Cy-Flaming
2	বর্তমানে কিছু নাটক তৈরি হচ্ছে যেগুলোর ভাষা ব্যবহার অশ্লীলতায় ভরা (Currently, some dramas are being made which are filled with obscene language)	Cy-Flaming
3	খুনের বদলে খুন হবে (Murder instead of murder)	Cy-Threat
4	আমার আইডি আনব্লক না করলে তোকে কেন তোর বাপের পা ও ভাঙবো (If you don't unblock my ID, I'll break your father's legs too)	Cy-Threat
5	আরেহ আফ্রিকার কালাদের মত লাগে ওরে (Looks like those African people, oh)	Cy-Racism
6	জাপানিদের আসল রূপ বের হয়ে আসছে এইবার (The true form of the Japanese is emerging this time)	Cy-Racism
7	তোর চেহারা দেখলেই আমার হাসি পায় (Just by looking at your face, I start laughing)	Cy-Pull-a-Pig
8	তোর চেহারা দেখলেই মেজাজটা কেমন যেন খারাপ হয়ে যায় (How bad the mood becomes just by looking at your face)	Cy-Pull-a-Pig
9	ভাই আপনার সঠিক সুন্দর আলোচনার জন্য অনেক অনেক ধন্যবাদ (Brother, thank you very much for your correct and beautiful conversation.)	Not Bullying
10	পুরোনো নাটক কিন্তু অনেক ভালো লাগে দেখলেই (Old dramas are still very enjoyable to watch)	Not Bullying

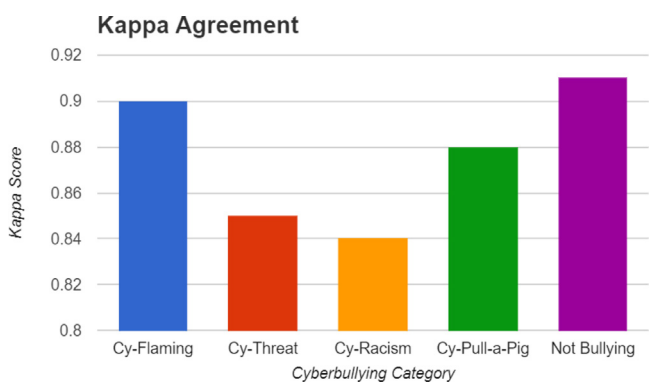

Fig. 5. Cohen's kappa scores of the classes.

the learning rate of 1e-5 is used for m-BERT, XLM-R and 1e-6 is used for B-BERT which determines the step size for updating the parameters during training while the dropout rate prevents overfitting by omitting neurons randomly. Dropout rate of 0.1 is used for B-BERT, XLM-R, and 0.5 is used for m-BERT. In the following sections, we will provide explanations for each model:

3.5.1. ML based models

In machine learning models, the dataset has been split into two parts: training (80%) and testing (20%).

- **Random forest:** A popular machine learning method that combines different decision trees to produce predictions.
- **Support vector machine (SVM):** SVM operates by locating an ideal hyperplane with the maximum margin of separation between data points belonging to distinct classes. It can solve both linear and non-linear classification issues using various kernel functions.
- **Multinomial Naive Bayes (MNB):** MNB is a probabilistic classification algorithm based on Bayes' theorem. It estimates the class with the highest probability after computing the probabilities for each class using the formula:

$$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)} \tag{1}$$

3.5.2. DL based models

In deep learning and transformer-based models, the dataset has been split into three parts: training (80%), testing (10%), and validation (10%).

- **Gated recurrent unit (GRU):** GRU is a specific style of recurrent neural network (RNN) architecture designed to address the vanishing gradient problem and identify long-term dependencies in sequential data
- **Convolutional neural network (CNN):** CNN is a deep learning model created for processing and interpreting structured grid-like data, such as photos and videos, but it can also be implemented for texts
- **Long short-term memory (LSTM):** LSTM architectures are designed to capture long-term dependencies in sequential data and overcome the vanishing gradient problem
- **Bidirectional LSTM (BiLSTM):** BiLSTM is a type of recurrent neural network that processes input sequences in both forward and backward directions simultaneously

For getting optimal performance of any deep learning or machine learning algorithm on a specific task, the hyperparameters needed to be carefully optimized. Description of the hyper-parameters tuned for our work are given below:

- **Epoch:** One whole run through the whole training dataset is referred to as an epoch. The model can learn more effectively by repeatedly iterating over the data across multiple epochs
- **Batch size:** The number of training samples processed in a batch before the model's internal parameters, or weights, are changed is known as the batch size. Although they demand more work every epoch, smaller batches offer more updates
- **Optimizer:** The optimizer algorithm modifies the weights in the model in order to minimize the loss function
- **Activation:** The model gains non-linearity from activation functions, which enables it to pick up intricate patterns. Tanh, ReLU, and Sigmoid are a few examples
- **Embedding dimension:** The embedding dimension in natural language processing describes the size of the dense vector that tokens or words are represented by. It establishes the word embeddings' feature space

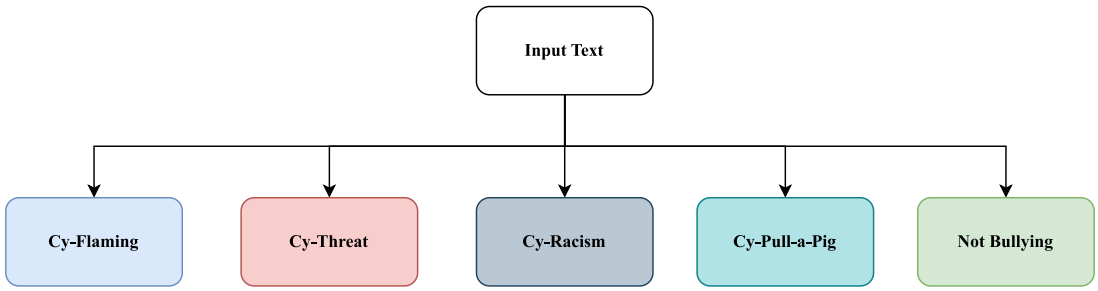

Fig. 6. Classification schema.

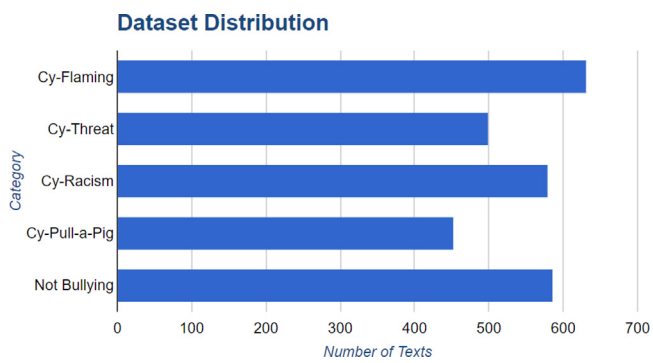

Fig. 7. Distribution of data.

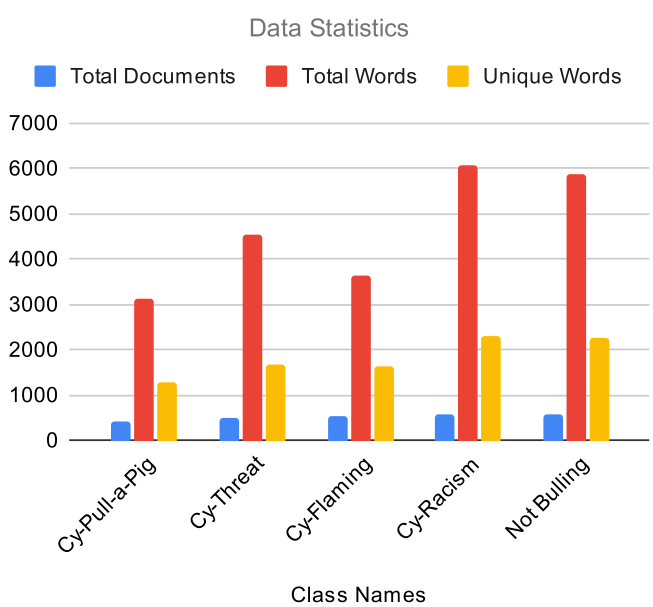

Fig. 8. Statistics of text data.

- **Dropout rate:** In order to avoid overfitting, dropout is a regularization technique that randomly changes a portion of input units to zero during training. The percentage of neurons lost is known as the dropout rate
- **Random state:** Reproducibility is ensured by the random state or seed, which manages the production of random numbers throughout procedures like data division and weight initialization
- **Maxpool:** In convolutional neural networks (CNNs), maxpooling, also known as down-sampling, is a down-sampling procedure that takes the maximum value over a sliding window to minimize the spatial dimensions of the input. This helps to preserve significant features while lowering computational load

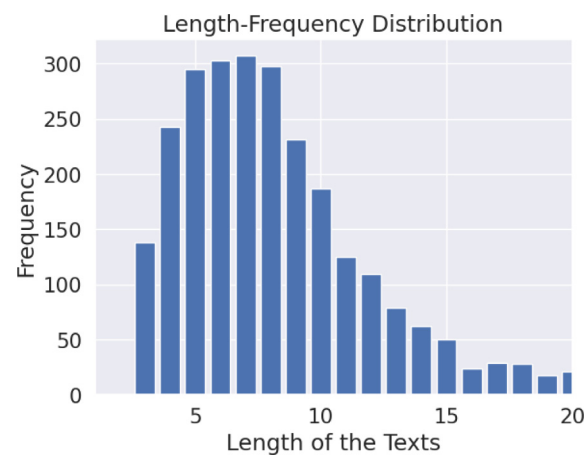

Fig. 9. Variation and frequency of length of texts.

Table 3 shows the optimized hyperparameters for machine learning and deep learning models.

3.5.3. *Transformers*

Transformers (Vaswani et al., 2017) are based on a self-attention mechanism that examines every segment of the sequence at once, in contrast to more antiquated architectures such as Recurrent Neural Networks (RNNs), which process sequences one step at a time. This is accomplished by each element in the sequence attending to (focusing on) other elements in order to comprehend their links and contextualize its own meaning which is known as self-attention.

- **Multilingual-BERT (m-BERT):** m-BERT (Devlin et al., 2018), Pires et al. (2019) is a transformer model pre-trained on a large corpus of multilingual data using unsupervised learning.
- **Bangla-BERT:** Bangla-BERT (Kowsher et al., 2022) is a pre-trained representation of the Bengali language that utilizes the mask language modeling technique.
- **XLM-RoBERTa:** XLM-RoBERTa (Conneau et al., 2019) is a version of RoBERTa that supports multiple languages and has been trained on a large corpus of data containing 100 different languages.

The hyperparameters optimized for fine tuning the transformer models are outlined in Table 4. The values of hyperparameters are taken from the ranges in respective "Hyperparameter space".

4. Result and evaluation

We have split our dataset and implemented several models. Pre-trained transformer models offer a very high scale performance when compared to machine learning, word embedding, and deep learning techniques when applied to the heavily context-dependent corpus (Kumar et al., 2020; Mandl et al., 2020; Anon, 2022). We have employed

Table 3
Hyperparameters values for machine learning and deep learning models.

Category	Hyperparameters
Machine Learning	
Random Forest (RF)	Random State = 95, Criterion = Gini, Min Samples Split = 2, Max Features = n_features
Support Vector Machine (SVM)	Random State = 95 , Kernel = Linear
Multinomial Naive Bayes (MNB)	Random State = 95 , Alpha = 1
Deep Learning	
Gated Recurrent Unit (GRU)	Epoch = 50, Batch Size = 32, Optimizer = Adam, Activation = Relu, Embedding Dimension = 64, Dropout = 0.2
Convolutional Neural Network (CNN)	Random State = 42, Epoch = 15, Batch Size = 64, Activation = Softmax, Dropout = 0.5, Maxpool = 2
Long Short-Term Memory (LSTM)	Random State = 68, Epoch = 35, Batch Size = 64, Activation = Softmax, Embedding Dimension = 64, Dropout = 0.5
Bidirectional LSTM (BiLSTM)	Random State = 95, Epoch = 35, Batch Size = 64, Activation = Softmax, Embedding Dimension = 64, Dropout = 0.5

Table 4
Fine tuning for transformers.

Name	Hyparameter space	m-BERT	B-BERT	XLM-R
Epoch Size	5, **7**, 9, **11**, 13, 16, **17**, 18	11	17	7
Batch Size	**2**, 4, 8, 16, 32	2	2	2
Learning Rate	1e−4, **1e-5**, **1e-6**, 1e−7	1e−5	1e−6	1e−5
Dropout Rate	**0.1**, 0.2, 0.4, **0.5**, 0.6, 0.7	0.5	0.1	0.1

Table 5
Execution time of different models.

Method	Number of epochs	Execution time
MNB	–	4 s
RF	–	11 s
SVM	–	3 s
BiLSTM	35	2 min 20 s
LSTM	35	75 s
CNN	15	34 s
GRU	50	26 s
XML-R	7	33 min 42 s
Bangla-BERT	17	72 min 2 s
m-BERT	11	48 min 13 s

various deep learning, machine learning, and transformer approaches. Additionally, we have utilized specific performance metrics and ROC curves to evaluate the models.

4.1. Evaluation matrices

The proposed textual cyberbullying model is assessed using a number of statistical metrics:

- **Confusion matrix:** A confusion matrix is a summary table that compares expected and actual class labels to assess how well a classification model performs.
  True positive (TP): The number of documents that are correctly predicted by the models that were positive.
  True negative (TN): The number of documents that are correctly predicted by the models that were negative.
  False positive (FP): The number of documents that are not predicted correctly which was actually positive.
  False negative (FN): The number of documents that are not predicted correctly which was actually negative.
- **Precision:** In binary and multi-class classification problems, precision is a performance statistic that is used to assess how accurately a model predicts positive outcomes.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

- **Recall:** Recall, often referred to as **sensitivity** or the **true positive rate**, quantifies the percentage of real positive occurrences that are accurately identified by a model, demonstrating the model's capacity to detect all positive instances.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

- **F1 score:** F1 score is the weighted average of Precision and Recall. It is the harmonic mean of precision and recall, combining both measures into a single value.

$$\text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{4}$$

- **Accuracy:** A classification model's overall correctness is determined by its accuracy. It is calculated by dividing the number of correct predictions by the total number of predictions.

$$\text{Accuracy} = \frac{TP + TN}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{5}$$

- **ROC curve:** The trade-off between the true positive rate and false positive rate for a predictive model utilizing various probability thresholds is summarized by Receiver Operating Characteristic (ROC) Curves. The threshold values provided by ROC curves range from 0 to 1.

4.2. Experimental setup and evaluation measures

Experiments conducted on the Google Collaboratory Platform using the GPU (Google Cloud Engine) backend in Python 3.10. To execute the models, 8 GB of disc space and 13 GB of RAM have been used. We processed and prepared the data using numpy (1.22.4) and pandas (1.5.3). While the DL models are trained using Keras (2.12.0) and TensorFlow (2.12.0), the machine learning models are constructed using the scikit-learn (1.2.2) tools. Models of transformers are created using the ktrain (0.32.1) packages (Maiya, 2022).

The Table 5 refers to the execution time of different machine learning, deep learning, and transformer models on this experimental setup. Although the pre-trained transformer models perform better their execution time is much higher than other models. We have also used early stopping so that the models do not get over-fitted.

4.3. Evaluation results

The Table 6 shows the weighted average of our models of machine learning, deep learning, and transformers. Here, in ML models MNB has the highest accuracy of 63.88% which is equivalent to 64% and it also has the highest precision, recall, and f1-score which is 0.64 for DL models BiLSTM has the highest accuracy of 74.59% and among all of the DL models it also has the highest precision of 0.75, recall of 0.75 and f-1 score of 0.74 and finally for the state-of-the-art language models like the transformers called XLM-RoBERTa gave us the most accurate result among all the ML, DL models and transformers that we have used which is 82.61% which also gave us the highest precision of 0.82, recall of 0.84 and f-1 score of 0.83 which is almost same as its accuracy. It can typically mean that our dataset is balanced and the model is capable of balancing precision and recall too.

Table 6
Results of ML, DL models and transformers.

Types	Algorithms	Precision	Recall	F1-Score	Accuracy
ML	RF	0.59	0.56	0.57	56.44%
	SVM	0.62	0.60	0.61	60.44%
	MNB	0.64	0.64	0.64	63.88%
DL	GRU	0.73	0.72	0.73	72.73%
	CNN	0.70	0.70	0.70	70.05%
	LSTM	0.75	0.74	0.74	74.05%
	BiLSTM	0.75	0.75	0.74	74.59%
Transformer	m-BERT	0.76	0.76	0.76	76.09%
	Bangla-BERT	0.82	0.80	0.81	81.16%
	XLM-R	**0.82**	**0.84**	**0.83**	**82.61%**

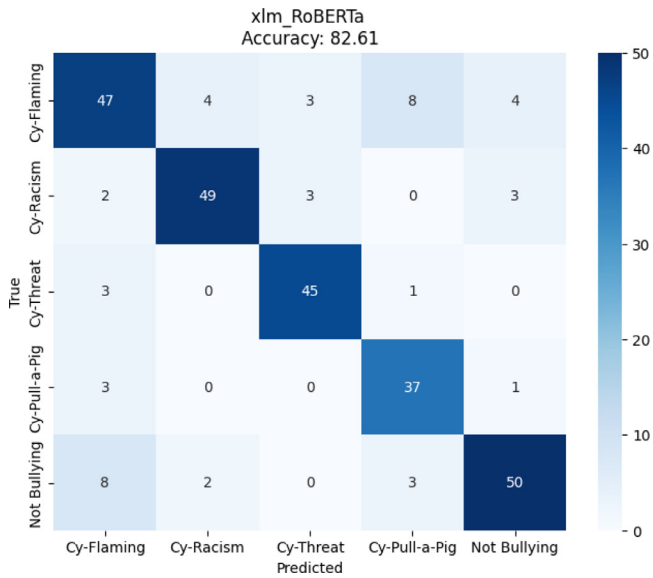

Fig. 10. Confusion matrix of XLM-RoBERTa.

4.4. Error analysis

In our work, Fig. 10 shows that XLM-RoBERTa has the largest amount of texts that they have predicted correctly with respect to true labels. For starters, we can say that among cyberbullying classes Cy-Racism has the highest true positive values where it classified 49 texts correctly and misclassified 8 texts.

Likewise, true positive values for Cy-Flaming, Cy-Threat, and Cy-Pull-a-Pig are 47, 45, and 37 among 66, 49, and 41 texts. Not Bullying class has classified 50 instances correctly and misclassified 13 instances. To conclude we must say that it is by far a better performing model for us than the machine learning and deep learning models we have used.

4.5. Discussion

The Fig. 11 is depicted to show the ROC curve for XLM-RoBERTa which gave us the best possible result. If the curve is steeper then it indicates better performance as the ROC indicated a true positive rate with respect to a false positive rate which means if the curve is steeper then the true positive rate is more prominent. In our work, we got the steepest curve for the XLM-RoBERTa and it has the highest AUC covered which means it has the most accurate performance among all the models that we have used.

In our study, we have employed three transformer models m-BERT, Bangla-BERT, and XLM-RoBERTa. The T5 (Raffel et al., 2020) is a powerful and potential transformer-based language model for handling text-to-text NLP tasks. T5 is trained on a massive dataset of text and code, the model learns valuable language representations that can be applied to new tasks with minimal training data. While our current

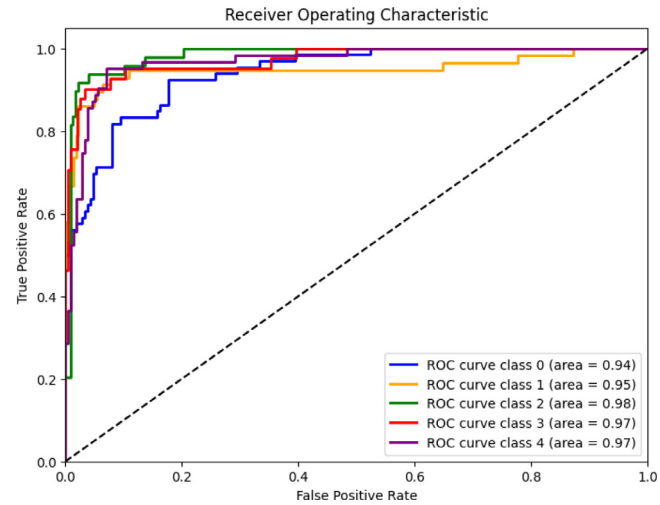

Fig. 11. ROC curve of XLM-RoBERTa.

study focuses on cyberbullying detection which is a text classification problem, the T5 model and its variants are primarily designed for tasks such as code generation (Gong et al., 2024), translation (Zhao et al., 2024). We recognize the potential of these models like prompt-Tuning (Huanzhen, 2023) to stimulate future research in cyberbullying detection.

4.6. Comparison with existing works

The evaluation results revealed that our proposed XLM-R model proved to be the most effective in cyberbullying detection. We compared the model's performance against several existing techniques to further assess its effectiveness. We implemented and evaluated several existing techniques on our developed dataset. The comparison results are presented in Table 7. The Wahid and Al Imran (2023) study obtained F1-score of 0.74 and accuracy of 74% while Tapaopong et al. (2024) study obtained F1-score of 0.76 and accuracy of 76%. The Nath et al. (2024) study obtained F1-score of 0.74 and accuracy of 74% by implementing two layer Bi-LSTM on our Bengali dataset whereas, in Shanto et al. (2023) they have proposed GRU which gave the F1-score of 0.73 and accuracy of 72.73%. Our work XLM-R outperforms every state-of-the-art works with 0.83 F1-score and 82.61% of accuracy.

5. Conclusion

The present study involved the creation of an extensive dataset on cyberbullying in Bengali, comprising 2751 texts, of which 2164 were classified as cyberbullying and 587 as non-bullying. The dataset serves as a fundamental resource for next studies and is further annotated into four distinct categories: Cy-Flaming, Cy-Threat, Cy-Pull-a-Pig, and Cy-Racism. To make sure the dataset was of high quality, we carried out in-depth data analysis. Furthermore, we fine-tuned the XLM-R model to achieve better performance than other models in identifying and classifying texts that engage in cyberbullying. By contrasting the performance of the suggested model with different machine learning, deep learning, and existing approaches, our work also creates a benchmark. We have unveiled the remarkable potential of transformer-based methods, notably exemplified by XLM-RoBERTa. Lastly, we conducted a thorough error analysis that yielded recommendations for further updates as well as qualitative and quantitative insights into the model's shortcomings.

With an impressive accuracy rate of 82.61% and an F1-score of 0.83, XLM-RoBERTa emerges as a frontrunner in effectively detecting instances of cyberbullying within Bengali text datasets. Our in-depth

Table 7
Comparison of performance with existing techniques.

Approach	Precision	Recall	F1-Score	Accuracy
Gated Multi-Feature BERT with Word2Vec (Wahid and Al Imran, 2023)	0.75	0.74	0.74	74%
Fine-tuned DistilBERT (Tapaopong et al., 2024)	0.77	0.76	0.76	76%
Two Layer Bi-LSTM (Nath et al., 2024)	0.75	0.74	0.74	74%
GRU (Shanto et al., 2023)	0.73	0.72	0.73	72.73%
Proposed	**0.82**	**0.84**	**0.83**	**82.61%**

comparative analysis, informed by insights gleaned from pertinent literature, underscores the robustness and efficacy of XLM-RoBERTa, advocating for its adoption as the preferred model for Bengali cyberbullying classification systems.

Through our study, we not only contribute to the advancement of cyberbullying detection in Bengali but also lay a solid foundation for future research endeavors in this critical area. By leveraging state-of-the-art techniques and datasets like the CBD, we can empower researchers and practitioners alike to combat cyberbullying more effectively and foster safer online environments for all users.

Based on the findings from the study referenced in Anon (2023), it is evident that XLM-RoBERTa outperforms both m-BERT and Bangla-BERT across various scenarios. Specifically, it has demonstrated superior performance compared to Bangla-BERT for our dataset. Therefore, we recommend opting for XLM-RoBERTa as the preferred model for our classification system in cyberbullying detection. This recommendation stems from its ability to achieve higher accuracy and effectiveness in handling the nuances of our dataset, making it a suitable choice for enhancing cyberbullying detection efforts in our field.

CRediT authorship contribution statement

Syed Sihab-Us-Sakib: Writing – original draft, Visualization, Resources, Methodology, Investigation, Data curation. **Md. Rashadur Rahman:** Writing – review & editing, Validation, Software, Resources, Methodology, Investigation, Formal analysis. **Md. Shafiul Alam Forhad:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Md. Atiq Aziz:** Writing – review & editing, Visualization, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abdullah, S., Forhad, M.S.A., Islam, M.S., Rahman, M.O., 2023. Classification of politically aggressive bengali texts using transformers. In: 2023 26th International Conference on Computer and Information Technology. ICCIT, IEEE, pp. 1–6.

Aboujaoude, E., Savage, M.W., Starcevic, V., Salame, W.O., 2015. Cyberbullying: Review of an old problem gone viral. J. Adolesc. Health 57 (1), 10–18.

Ahmed, M.T., Rahman, M., Nur, S., Islam, A., Das, D., 2021. Deployment of machine learning and deep learning algorithms in detecting cyberbullying in bangla and romanized bangla text: A comparative study. In: 2021 International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies. ICAECT, IEEE, pp. 1–10.

Ahmed, M.T., Rahman, M., Nur, S., Islam, A.T., Das, D., 2022. Introduction of pmi-so integrated with predictive and lexicon based features to detect cyberbullying in bangla text using machine learning. In: Proceedings of 2nd International Conference on Artificial Intelligence: Advances and Applications: ICAIAA 2021. Springer, pp. 685–697.

Akhter, A., Acharjee, U.K., Talukder, M.A., Islam, M.M., Uddin, M.A., 2023. A robust hybrid machine learning model for bengali cyber bullying detection in social media. Natural Lang. Process. J. 4, 100027.

Anon, 2022. BERT 101. Online; https://huggingface.co/blog/bert-101. (Accessed 17 May 2023).

Anon, 2023. XLM-roberta. https://huggingface.co/docs/transformers/model_doc/xlm-roberta. (Accessed 25 June 2023).

Aurpa, T.T., Sadik, R., Ahmed, M.S., 2022. Abusive bangla comments detection on facebook using transformer-based deep learning models. Soc. Netw. Anal. Min. 12 (1), 24.

Bello, A., Ng, S.-C., Leung, M.-F., 2023. A BERT framework to sentiment analysis of tweets. Sensors 23 (1), 506.

Bender, E.M., Friedman, B., 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. Trans. Assoc. Comput. Linguist. 6, 587–604.

Bozyiğit, A., Utku, S., Nasibov, E., 2021. Cyberbullying detection: Utilizing social media features. Expert Syst. Appl. 179, 115001.

Chakraborty, P., Seddiqui, M.H., 2019. Threat and abusive language detection on social media in bengali language. In: 2019 1st International Conference on Advances in Science, Engineering and Robotics Technology. ICASERT, IEEE, pp. 1–6.

Chelmis, C., Yao, M., 2019. Minority report: Cyberbullying prediction on instagram. In: Proceedings of the 10th ACM Conference on Web Science. pp. 37–45.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., Stoyanov, V., 2019. Unsupervised cross-lingual representation learning at scale. arXiv preprint arXiv:1911.02116.

Das, A.K., Ashrafi, A., Ahmmad, M., 2019. Joint cognition of both human and machine for predicting criminal punishment in judicial system. In: 2019 IEEE 4th International Conference on Computer and Communication Systems. ICCCS, IEEE, pp. 36–40.

Das, A., Sharif, O., Hoque, M.M., Sarker, I.H., 2021. Emotion classification in a resource constrained language using transformer-based approach. arXiv preprint arXiv:2104.08613.

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.

Dewani, A., Memon, M.A., Bhatti, S., 2021. Cyberbullying detection: advanced preprocessing techniques & deep learning architecture for Roman Urdu data. J. Big Data 8 (1), 1–20.

Dong, X., Yu, Z., Cao, W., Shi, Y., Ma, Q., 2020. A survey on ensemble learning. Front. Comput. Sci. 14, 241–258.

Emon, M.I.H., Iqbal, K.N., Mehedi, M.H.K., Mahbub, M.J.A., Rasel, A.A., 2022. Detection of bangla hate comments and cyberbullying in social media using nlp and transformer models. In: International Conference on Advances in Computing and Data Sciences. Springer, pp. 86–96.

Faisal Ahmed, M., Mahmud, Z., Biash, Z.T., Ryen, A.A.N., Hossain, A., Ashraf, F.B., 2021. Cyberbullying detection using deep neural network from social media comments in bangla language. arXiv e-prints, arXiv–2106.

Fortuna, P., da Silva, J.R., Wanner, L., Nunes, S., et al., 2019. A hierarchically-labeled portuguese hate speech dataset. In: Proceedings of the Third Workshop on Abusive Language Online. pp. 94–104.

Gong, L., Elhoushi, M., Cheung, A., 2024. AST-T5: Structure-aware pretraining for code generation and understanding. arXiv preprint arXiv:2401.03003.

Hellfeldt, K., López-Romero, L., Andershed, H., 2020. Cyberbullying and psychological well-being in young adolescence: the potential protective mediation effects of social support from family, friends, and teachers. Int. J. Environ. Res. Public Health 17 (1), 45.

Hemmatian, F., Sohrabi, M.K., 2019. A survey on classification techniques for opinion mining and sentiment analysis. Artif. Intell. Rev. 52 (3), 1495–1545.

Huanzhen, L., 2023. Improve code summarization via prompt-tuning CodeT5. Wuhan Univ. J. Nat. Sci. 28 (6), 474–482.

Ishmam, A.M., Sharmin, S., 2019. Hateful speech detection in public facebook pages for the bengali language. In: 2019 18th IEEE International Conference on Machine Learning and Applications. ICMLA, IEEE, pp. 555–560.

Islam, S., Forhad, M.S.A., Murad, H., 2022. Banglasapm: A deep learning model for suicidal attempt prediction using social media content in bangla. In: 2022 25th International Conference on Computer and Information Technology. ICCIT, IEEE, pp. 1122–1126.

Kee, D.M.H., Al-Anesi, M.A.L., Al-Anesi, S.A.L., 2022. Cyberbullying on social media under the influence of COVID-19. Glob. Bus. Organ. Excell. 41 (6), 11–22.

Khan, U., Khan, S., Rizwan, A., Atteia, G., Jamjoom, M.M., Samee, N.A., 2022. Aggression detection in social media from textual data using deep learning models. Appl. Sci. 12 (10), 5083.

Kowsher, M., Sami, A.A., Prottasha, N.J., Arefin, M.S., Dhar, P.K., Koshiba, T., 2022. Bangla-BERT: transformer-based efficient model for transfer learning and language understanding. IEEE Access 10, 91855–91870.

Kumar, R., Ojha, A.K., Malmasi, S., Zampieri, M., 2020. Evaluating aggression identification in social media. In: Proceedings of the Second Workshop on Trolling, Aggression and Cyberbullying. European Language Resources Association (ELRA), Marseille, France, pp. 1–5, URL https://aclanthology.org/2020.trac-1.1.

Kumar, A., Sachdeva, N., 2022. A bi-GRU with attention and CapsNet hybrid model for cyberbullying detection on social media. World Wide Web 25 (4), 1537–1550.

Lu, N., Wu, G., Zhang, Z., Zheng, Y., Ren, Y., Choo, K.-K.R., 2020. Cyberbullying detection in social media text based on character-level convolutional neural network with shortcuts. Concurr. Comput.: Pract. Exper. 32 (23), e5627.

Mahmud, T., Das, S., Ptaszynski, M., Hossain, M.S., Andersson, K., Barua, K., 2022. Reason based machine learning approach to detect bangla abusive social media comments. In: International Conference on Intelligent Computing & Optimization. Springer, pp. 489–498.

Maiya, A.S., 2022. Ktrain: A low-code library for augmented machine learning. J. Mach. Learn. Res. 23 (158), 1–6.

Mandl, T., Modha, S., Kumar M, A., Chakravarthi, B.R., 2020. Overview of the hasoc track at fire 2020: Hate speech and offensive language identification in tamil, malayalam, hindi, english and german. In: Forum for Information Retrieval Evaluation. pp. 29–32.

Nath, S.S., Karim, R., Miraz, M.H., 2024. Deep learning based cyberbullying detection in bangla language. arXiv preprint arXiv:2401.06787.

Pascucci, A., Masucci, V., Monti, J., 2019. Computational stylometry and machine learning for gender and age detection in cyberbullying texts. In: 2019 8th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos. ACIIW, IEEE, pp. 1–6.

Pires, T., Schlinger, E., Garrette, D., 2019. How multilingual is multilingual BERT? arXiv preprint arXiv:1906.01502.

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., Liu, P.J., 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. J. Mach. Learn. Res. 21 (140), 1–67.

Ross, B., Rist, M., Carbonell, G., Cabrera, B., Kurowsky, N., Wojatzki, M., 2017. Measuring the reliability of hate speech annotations: The case of the european refugee crisis. arXiv preprint arXiv:1701.08118.

Sen, O., Fuad, M., Islam, M.N., Rabbi, J., Masud, M., Hasan, M.K., Awal, M.A., Fime, A.A., Fuad, M.T.H., Sikder, D., et al., 2022. Bangla natural language processing: A comprehensive analysis of classical, machine learning, and deep learning-based methods. IEEE Access 10, 38999–39044.

Shakambhari, Raj, J.S., Anantha Babu, S., 2022. Smart cyberbullying detection with machine learning. In: Disruptive Technologies for Big Data and Cloud Applications: Proceedings of ICBDCC 2021. Springer, pp. 237–248.

Shanto, S.B., Islam, M.J., Samad, M.A., 2023. Cyberbullying detection using deep learning techniques on bangla facebook comments. In: 2023 International Conference on Intelligent Systems, Advanced Computing and Communication. ISACC, IEEE, pp. 1–7.

Talpur, B.A., O'Sullivan, D., 2020. Cyberbullying severity detection: A machine learning approach. PLoS One 15 (10), e0240924.

Talpur, K.R., Yuhaniz, S.S., Amir, N., 2020. Cyberbullying detection: Current trends and future directions. J. Theor. Appl. Inf. Technol. 98, 3197–3208.

Tapaopong, W., Charoenphon, A., Raksasri, J., Samanchuen, T., 2024. Enhancing cyberbullying detection on social media using transformer models. In: 2024 5th Technology Innovation Management and Engineering Science International Conference. TIMES-ICON, IEEE, pp. 1–5.

Tokunaga, R.S., 2010. Following you home from school: A critical review and synthesis of research on cyberbullying victimization. Comput. Hum. Behav. 26 (3), 277–287.

Van Hee, C., Jacobs, G., Emmery, C., Desmet, B., Lefever, E., Verhoeven, B., De Pauw, G., Daelemans, W., Hoste, V., 2018. Automatic detection of cyberbullying in social media text. PLoS One 13 (10), e0203794.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. In: Advances in Neural Information Processing Systems, vol. 30.

Vidgen, B., Derczynski, L., 2020. Directions in abusive language training data, a systematic review: Garbage in, garbage out. Plos one 15 (12), e0243300.

Wahid, Z., Al Imran, A., 2023. Multi-feature transformer for multiclass cyberbullying detection in bangla. In: IFIP International Conference on Artificial Intelligence Applications and Innovations. Springer, pp. 439–451.

Willard, N.E., 2007. Cyberbullying and cyberthreats: responding to the challenge of online social aggression, threats, and distress. Research press.

Zhao, S., Tian, J., Fu, J., Chen, J., Wen, J., 2024. FeaMix: Feature mix with memory batch based on self-consistency learning for code generation and code translation. IEEE Trans. Emerg. Top. Comput. Intell..